

Processing of Metals 1

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Aims of session

Present

Present a classification of metal processing techniques



Describe

Describe some of the common fabrication techniques for metals



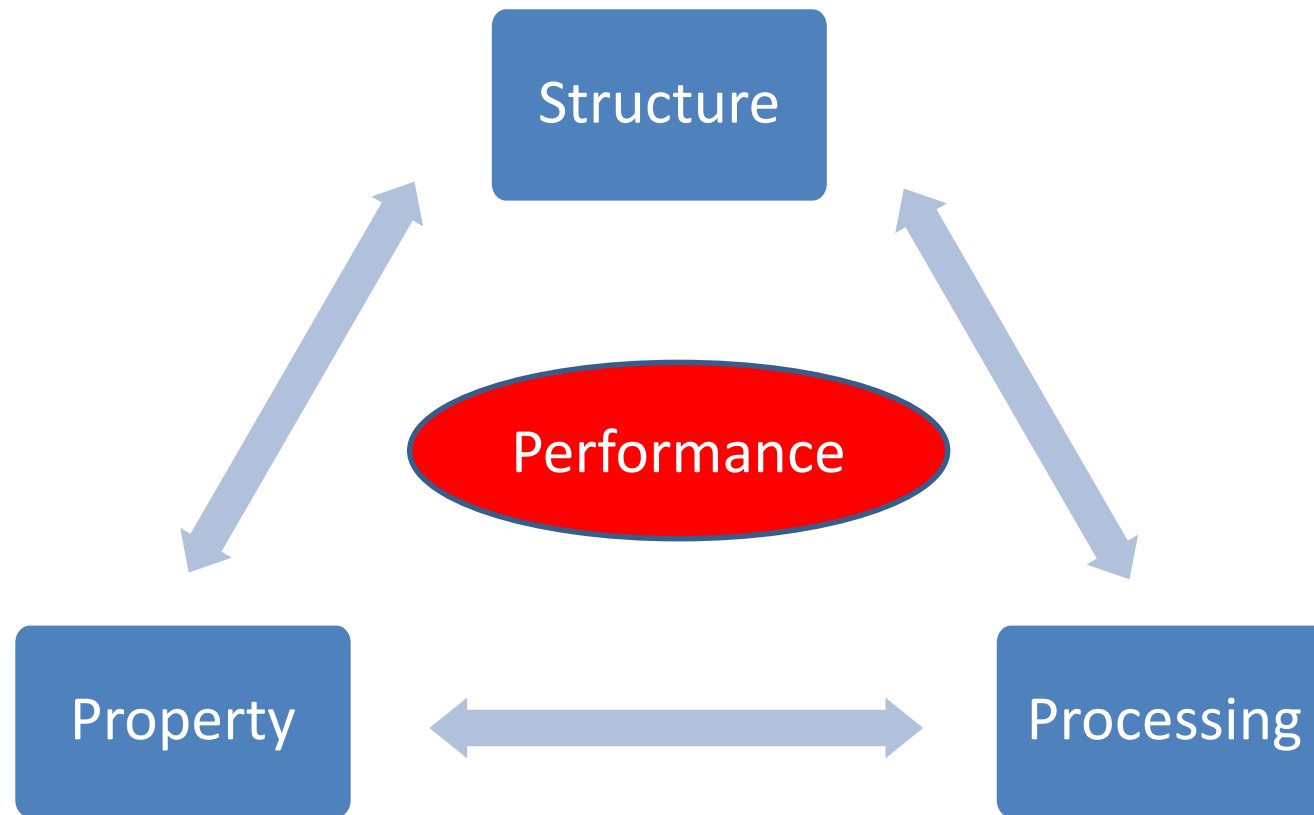
Explain

Explain what heat treatment procedures are used to improve the mechanical properties of both ferrous and non ferrous alloys

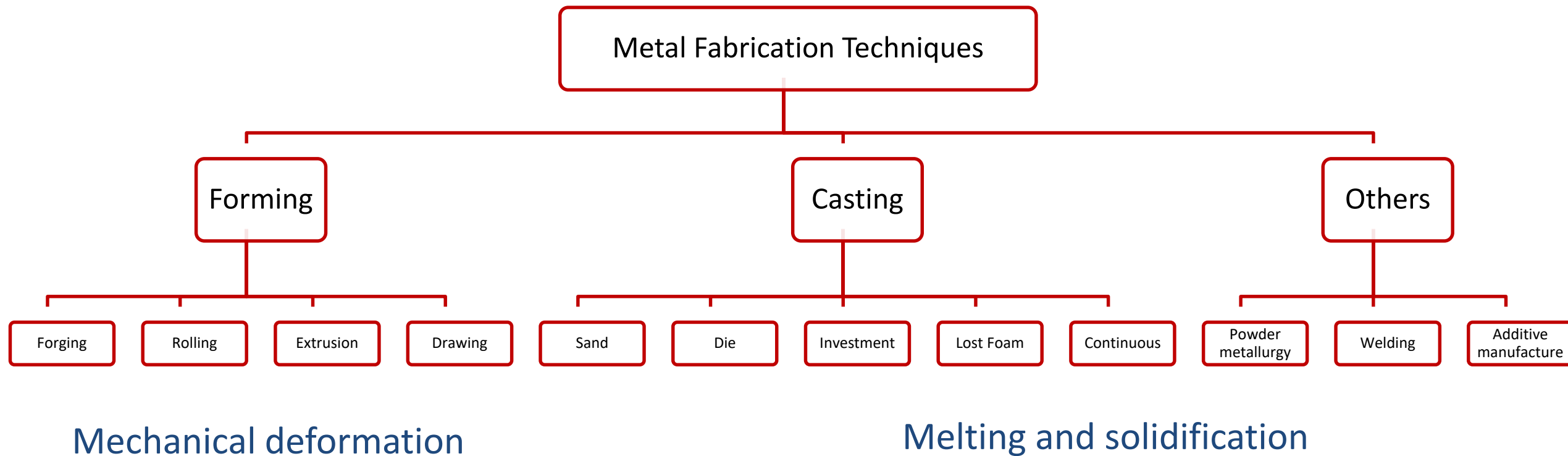
Foundation of Materials Engineering



When we process the material, we change the structure and the properties also change



Classification of metal processing techniques :



Forming operations

Most metals begin by casting a large block (**billet**)

Wrought metal alloys can then be shaped by plastic deformation

Forming processes:

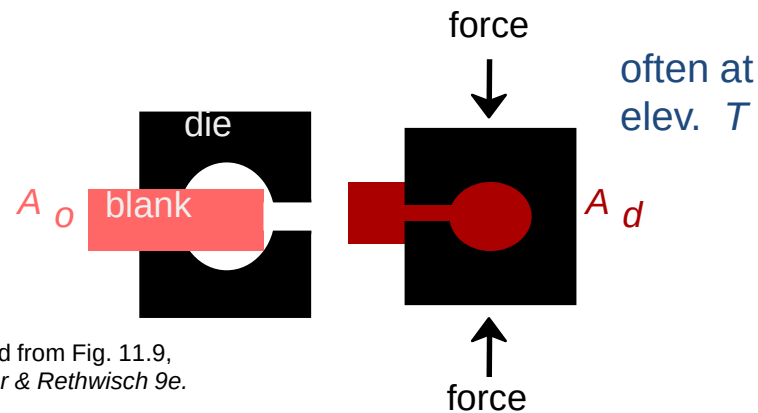
- Forging
- Cold Rolling
- Hot rolling
- Extrusion
- Drawing



All of these will change the microstructure of the metal!

Forging

Forging is the hammering or stamping of metal parts
(wrenches, crankshafts, car body panels)

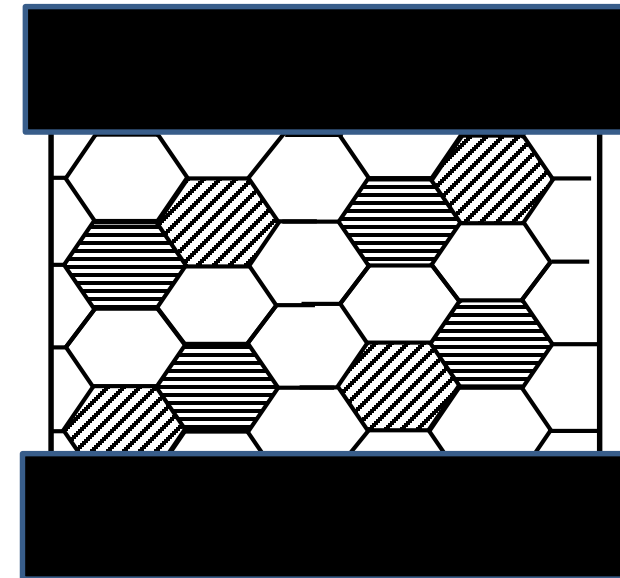


Adapted from Fig. 11.9,
Callister & Rethwisch 9e.

- Metal billet is pressed into shape using a hammer (forge) or press tool.
- **Open forge** = flat plates
- **Closed forge** = shaped to the component
- Metal is usually heated to deform easily
- May need to be re-heated several times

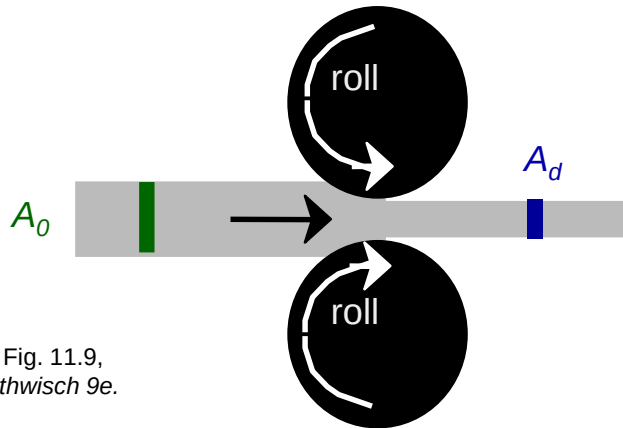
Deformation will change the grain shape

Hot forging = dynamic recovery of crystal shape (rapid annealing)



Rolling

Rolling is the pressing of the metal between large rollers
The rollers may be smooth or shaped to the section that is required
(sheet and plate, I beams and rails)



Adapted from Fig. 11.9,
Callister & Rethwisch 9e.



$$\% \text{ reduction} = A_d/A_0$$

This depends of the deformation the metal can stand.

Cold rolling:

Below the recrystallization temperature

Microstructure will be elongated grains

High degree of plastic strain

The metal may need to be annealed between passes

Hot rolling:

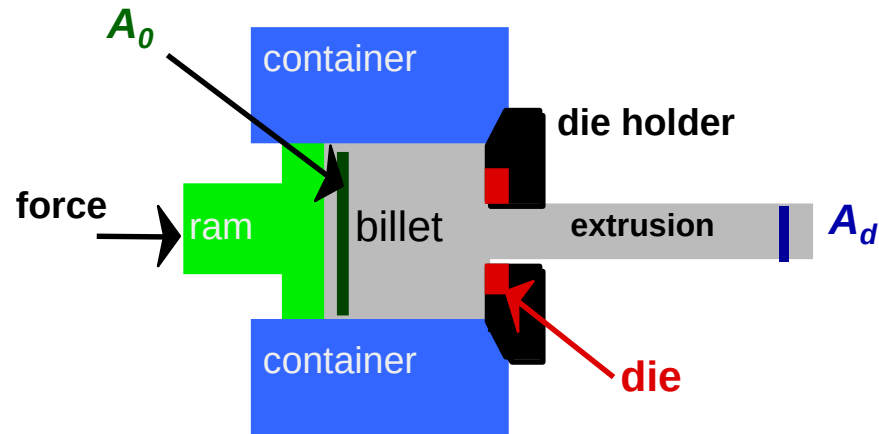
Above the recrystallization temperature

Dynamic recovery of grain structure

Large % reduction (deformation) per pass

Extrusion and drawing

Extrusion pushes the metal through a shaped die (e.g. rods and tubing, heat sinks)

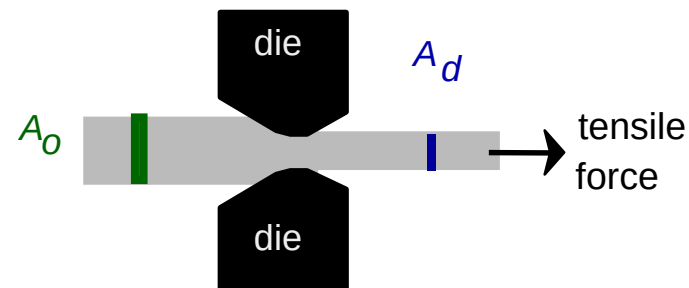


ductile metals, e.g. Cu, Al (hot)



Aluminium extrusions

Drawing pulls the metal through a shaped die (e.g. wire, rod, tubing)

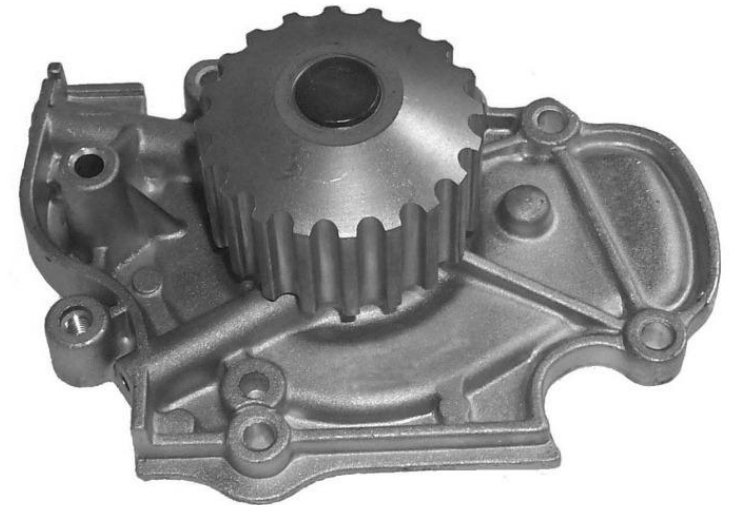


die must be well lubricated & clean

Casting

Casting for metals:

- Heat to above melting temperature (liquid)
- Pour into mould to make the shape you want
(mould material must be able to withstand temperature)
- Cool to room temperature – open the mould
- Most economical compared to other fabrication processes
- Gives good production of shapes
- Good option for brittle materials



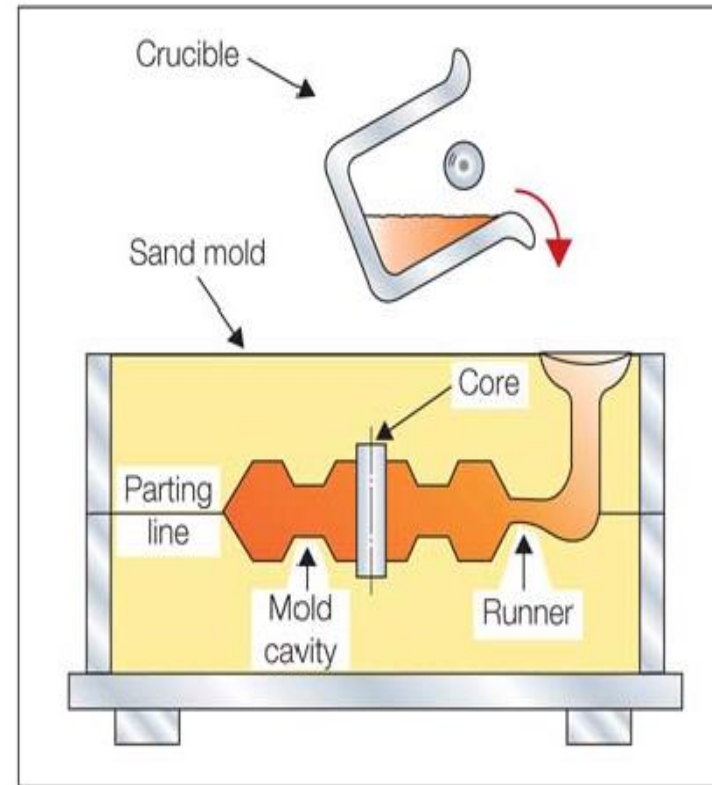
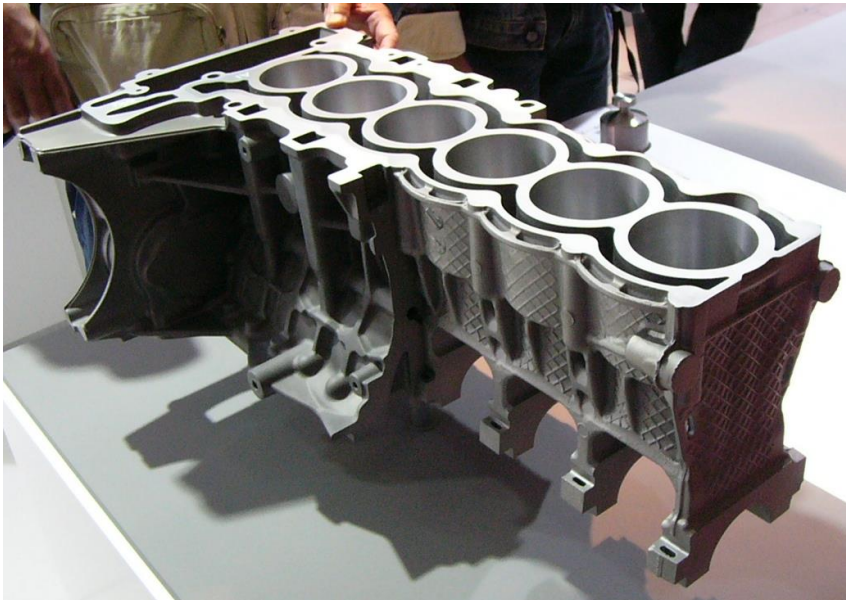
Casting

Casting for metals:

Sand Casting

(large parts, e.g., auto engine blocks)

Create the shape you want to mold
(**pattern**) and use to shape the sand



Sand casting

Casting for metals:

Investment Casting - lost wax process
(small complex shaped parts)

- Create the pattern in **wax**
(injection molded or now 3D printed wax)
- Cover in plaster (CaSO_4)
- Melt out the wax
- Pour in the molten metal
- Cool and break open the mold



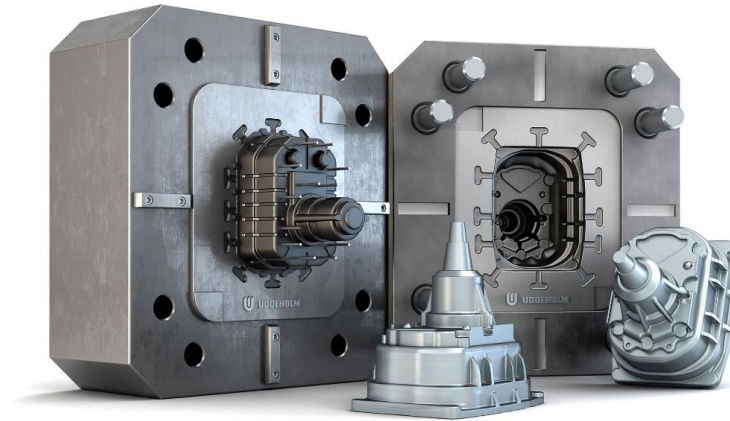
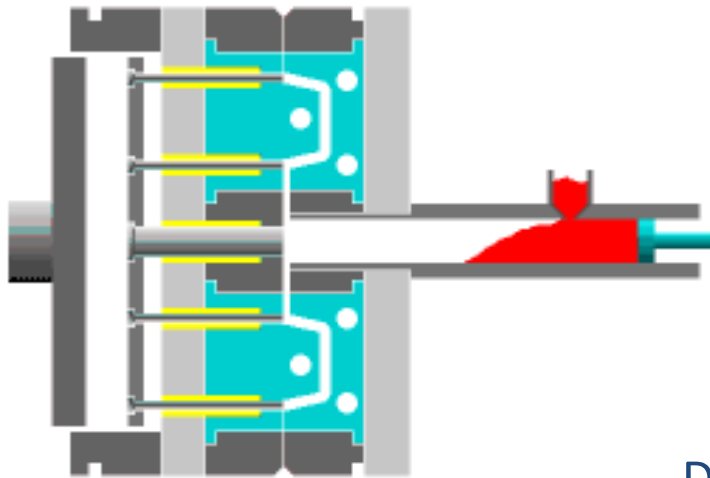
Casting

Casting for metals:

Die casting

(complex shaped parts)

- Inject molten metal into a metal mold
- Can make many parts from the same mold



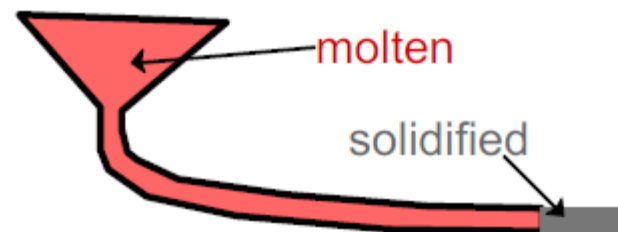
Die casting is good for lower T_m metals
Aluminium, Magnesium

Casting for metals:

Continuous casting

simple shaped parts (e.g., rectangular slabs, cylinders)

- Metal is heated until it liquefies
- Molten metal is then allowed to solidify until it becomes a semi-finished slab
- The molten metal is continuously supplied to the mold



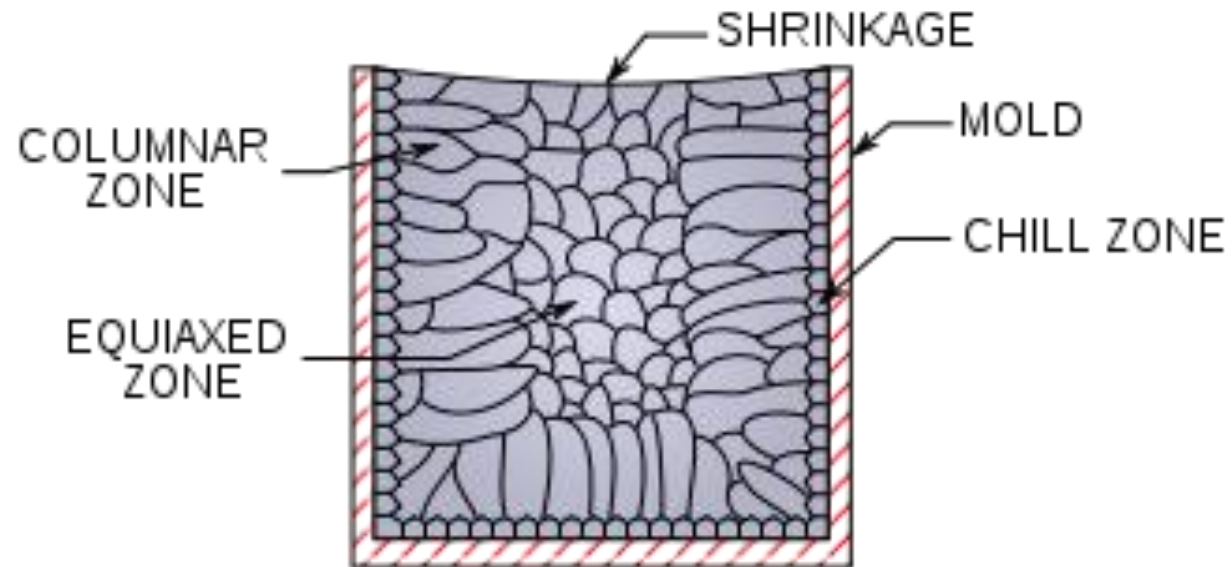
Casting

Casting for metals:

Cast microstructure

Fast cooling at the surface = small grains

Slow cooling in the middle = equiaxed grains



Welding

Welding of metals:

Welding is a **local casting process**

Melting and solidification of metal to form a joint

Rapid cooling = small grain size

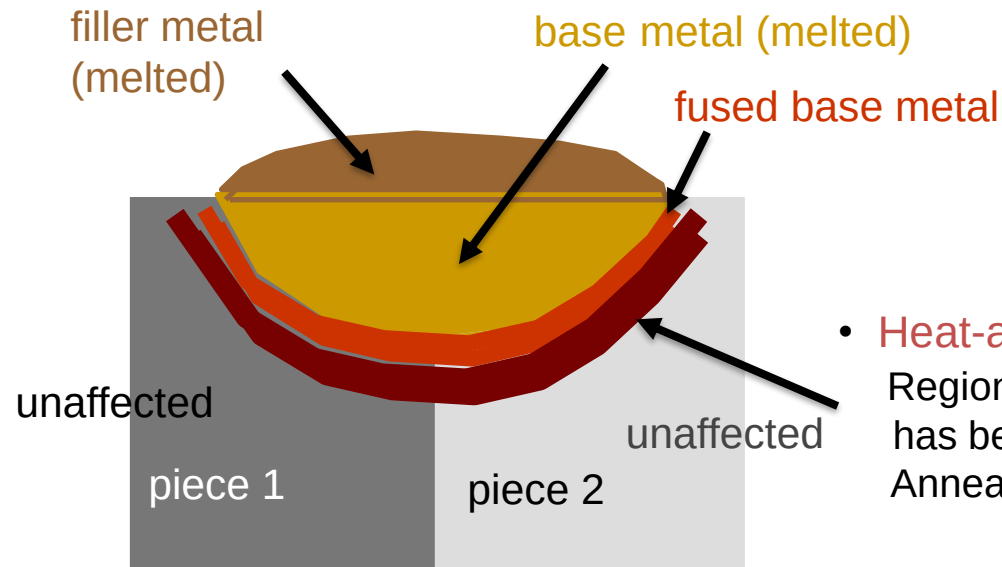


Photo: wnjay_woothisak/iStockphoto

- **Heat-affected zone:**
Region in which the microstructure has been changed but melted.
Annealed base metal = coarse grain size

Powder metallurgy

Sintering of metals:

Fusing of metal powder below the melting temperature

Joining of powder particles by surface diffusion = heat to $\sim 0.8T_m$

Good for metals that are difficult to cast and the **refractory metals**

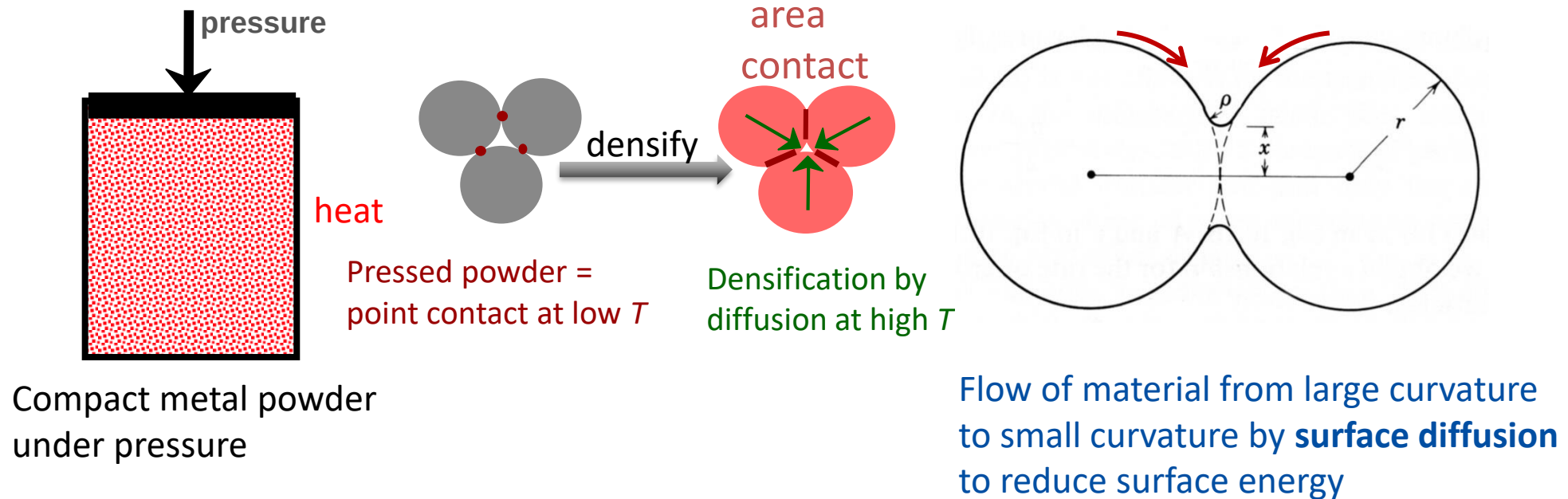


Fig. 11.10, *Callister & Rethwisch 9e*.
[From *Iron Castings Handbook*, C.F. Walton and T.J. Opar (Ed.),
Iron Castings Society,
Des Plaines, IL, 1981.]

Metal Additive Manufacturing

Powder based metal additive manufacturing **Selected laser melting (SLM)**

Local melting of the metal powder (casting compositions)
Fused into solid only where the laser heats it
Local cast metal = very small volume
Melt pool size and freezing rate of metal important



Loose powder clean up and
poor surface finish are disadvantages

Metal processing



Typical sequence of metal processing operations

- Cast billet of alloy
- Hot forge, roll or press to approximate shape
- Heat treat to develop microstructure and desired strength
- Finish machine to final shape
- Anneal to reduce internal (residual) stresses

Learning Outcomes

What can you do now?

- Select metals based on their (mechanical) properties for different applications
- Understand the difference between steel and cast iron
- Understand the different characteristics of non-ferrous metals
- Understand the processing methods for metal and the effect these will have on the microstructure
- Select processing methods and heat treatments to achieve the desired performance of metals